

Albert Zhang

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Experience

Insource Services

May 2025 – Present

Data Solutions Engineer

Wellesley, MA

- Built Tutorsim, the open-source benchmark runtime behind TutorMoments-Preview (first author; Ai2, MIT, Stanford): a Python CLI that runs tutor models against a simulated student over 520 expert-annotated moments and scores continuations through a 3-pass LLM classifier, with provider-agnostic routing across model APIs.
- Built and ran the evaluation pipeline benchmarking 11 VLMs on 56,000+ QA pairs over 2,030 images of real student handwritten math work, improving scoring reliability with a three-model majority-vote LLM judge validated against human ratings.
- Designed, built, and deployed to production a lead-discovery system on Azure (Django, Celery/Redis, Terraform, GitHub Actions CI/CD) that scores 14,000+ investment firms against a fund's criteria, turning raw regulatory filings into ranked prospects with generated fit briefs via LLM APIs.
- Architected an ELT pipeline in Snowflake, dbt, and Dagster, ingesting nested GraphQL API responses into landing tables and flattening them through layered dbt models into queryable marts, orchestrated end to end with event-triggered transformations.

Aramco

May 2024 – Aug. 2024

Research and Development Intern

Boston, MA

- Optimized a bio-polymer aerogel for carbon capture through iterative synthesis and testing, improving CO₂ capture efficiency 96% over baseline, and first-authored the resulting paper.

Research

TutorMoments-Preview: When Help is Unhelpful — Evaluating AI Tutors for Productive Struggle

A. Zhang, A. Ross, K. Patel, J. Bernado, R. Bowie, A. Trindade Ribeiro, D. Halper, H. Valayaputtur, J. Andreas, S. Loeb, L. Li, K. Lo, R. Knight. Preprint, 2026. [paper](#) · [site](#) · [code](#) · [data](#)

The Aftermath of DrawEduMath: Vision Language Models Underperform with Struggling Students and Misdiagnose Errors

L. Li, A. Zhang, N. Anderson, R. Knight, K. Lo. Preprint, 2026. Extends DrawEduMath (NAACL 2025 Outstanding Paper). [arXiv](#) · [site](#)

Projects

Compliance Analyzer | TypeScript, Next.js, Claude API, Docker

2026

- Built a document-audit system that checks internal policies against a regulatory source, sustaining 10-way concurrent scoring under API rate limits with prompt caching over the policy corpus.
- Batched corpora exceeding the model's context window and scored every requirement match with an evidence quote and a confidence level for reviewer override.

Blackbox | Python, Qdrant, Ollama, Docker

github.com/Bdfihn/Blackbox

- Built a self-hosted logging system that indexes PC/mobile activity, git history, and wearable data in a local Qdrant store and answers natural-language questions over it, running fully offline on Ollama.

Education

Boston University

Sep. 2021 – May 2025

B.A. Computer Science and B.A. Biochemistry & Molecular Biology

Boston, MA

Technical Skills

Languages: Python, SQL, TypeScript, Java, C/C++

ML & Evaluation: LLM/VLM evaluation (benchmark design, LLM-as-a-judge, BERTScore, rubric design), large-scale batch inference, RAG and vector search, LLM/VLM APIs (Anthropic, OpenAI, Google), scikit-learn

Backend & Data: FastAPI, Django, Celery, ARQ, PostgreSQL, Supabase, Qdrant, Redis, Snowflake, dbt, Dagster

Cloud & DevOps: AWS, Azure, GCP, Docker, Terraform, GitHub Actions, pytest

Certifications: Anthropic Certified Architect (2026), Certified Scrum Developer (2026–2028)